Dr. Anal Bhowmik

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Academic Qualification

- **Doctor of Philosophy (Ph.D.)** from Indian Institute of Technology Kharagpur, India in 2018.

  Thesis: Interaction of vortex beam with ultra-cold atoms: Mechanism and applications.
  Supervisor: Prof. Sonjoy Majumder
- **Master of Science** with First class from Indian Institute of Technology Roorkee, India in 2012.
  Thesis: Hubbard model for strongly correlated electronic system.
  Supervisor: Prof. Ishwar Singh
- **Bachelor of Science** with Second class from University of Burdwan, India in 2008.
- **Higher Secondary (12th)** with First division from West Bengal Council of Higher Secondary Education, India in 2005.
- **Madhyamik (10th)** with First division from West Bengal Board of Secondary Education, India in 2003.

Professional Experiences

- **Post-Doctoral Fellow** at University of Oklahoma, Oklahoma, USA. April 2023 to March 2026.
  Broad area of research:. Properties of Rydberg atom and its interaction with light.
  Mentor: Prof. Doerte Blume
- **Post-Doctoral Fellow** at University of Haifa, Haifa, Israel – December 2018 to March 2023.
  Broad area of research: 1. Many-body quantum dynamics and excitations of Bose-Einstein condensates in 2D traps.
  2. Coupled-cluster theory for trapped bosonic mixtures.
  Mentor: Prof. Ofir E. Alon

Scientific expertise

- **Light-matter interaction:** I am interested in the microscopic interaction of optical vortex with ultra-cold atomic system. Specifically, the transfer mechanisms of both orbital and spin angular momentum from optical vortex to Bose-Einstein condensate. The optical vortex could be paraxial, non-paraxial, and asymmetric in nature.
- **Atomic spectroscopy:** I work on the relativistic many-electron theory, namely, the relativistic coupled-cluster theory, to accurately find various atomic properties, to accurately find various atomic properties, such as, ionization potentials, excitation energies, isotopic shift, allowed and

forbidden atomic transition matrix elements, and hyperfine splitting for various atoms and ions.

- **Polarizability and magic wavelengths of ground state and low-lying states:** I am interested in the determination of the dynamic polarizability for atomic states subjected to external field of Gaussian laser and vortex laser. I use relativistic coupled cluster theory to accurately calculate the dynamic polarizability and magic wavelengths for the ground and low-lying atomic states.
- **Many-body theory of Bose Einstein condensate:** One of my research interests is the out-of-equilibrium tunneling dynamics in bosonic Josephson junctions (BJJ) by using the many-body level of theory. The many-body theory is correlation exhaustive and known as the multiconfigurational time-dependent Hartree method. Here, I investigate the fragmentations of bosons and its impact on the observable in the dynamics.
- **Developing and implementing coupled-cluster theory for bosonic systems:** The mean-field theory is the standard starting point for the coupled-cluster ansatz, atop which excitation operators are defined. The contraction embodied by the coupled-cluster ansatz is particularly suitable for bosons in traps. With just a few excitation operators, various correlations can be accurately captured for larger bosonic systems than currently available by other methods. More so, the coupled-cluster ansatz offers a natural connection between entanglement and excitations atop the mean-field solution. We have so far implemented the coupled-cluster theory for (identical bosons and) a mixture of two kinds of identical bosons in its most basic level, coupled-cluster (singles) doubles, with $M_1 = M_2 = 2$ orbitals.
- **Optical trapping of Rydberg atoms:** Electromagnetic fields fundamentally alter the optical properties of materials and are nowadays routinely used to trap cold atoms as well as microparticles and microspheres. My research interest is applying force on an atom with a high principal quantum number n, specifically an alkali Rydberg atom, in the presence of an external electromagnetic field. The trapping light could have linear and nonlinear polarization.

Undergraduate Student Mentoring

- **University of Oklahoma 2023 to present** Mentoring one undergraduate student in developing code for investigating the trapping of double valence Rydberg atom.

Graduate Student Mentoring

- **University of Oklahoma, 2024-present** Mentoring one graduate student within our research group, in developing the trapping of interacting Rydberg atoms with vortex laser.
- **Indian Institute of Technology, Kharagpur 2017-18** Assisted my supervisor in mentoring three graduate students within our research group, providing guidance in various areas of atomic and optical physics. One student focused on atomic spectroscopy in a plasma environment, another explored optical trapping of atoms and ions, and the third investigated the interaction between vortex light and Bose-Einstein condensates.

Post-doctoral fellow Mentoring

- **University of Haifa 2023 to present** Mentoring a postdoctoral fellow in research on the static and dynamic properties of a rotating Bose-Einstein condensate in a radial double well using the many-body method.

Teaching experience

- During my PhD, I taught undergraduate and master's students for five years (2013 to 2018). The subjects I taught included classical mechanics, quantum mechanics, and electromagnetic theory.
- I also took optics laboratory classes for undergraduate students.

Fellowships

- **Merit-cum-Means Fellowship** from Indian Institute of Technology Roorkee, India in 2011-2012.
- **Junior Research Fellowship (JRF)** from Indian Institute of Technology, Kharagpur, India in 2013-2014.
- **Senior Research Fellowship (SRF)** from Indian Institute of Technology, Kharagpur, India in 2015-2018.
- Post-Doctoral Fellows financed by the Council for Higher Education's Internationalization Project, Israel in 2018-2019.

Journal Referring

- Physical Review A, Journal of Chemical Physics, and Annals of Physics.

Awards and Memberships

- Joint Entrance Screening Test **(JEST-2009)** from Physics discipline.
- Joint Admission test for M.Sc. **(JAM-2010)** from Physics discipline.
- Graduate Aptitude Test **(GATE-2012)** from Physics discipline.
- Joint Council of Scientific & Industrial Research (CSIR)- University Grants Commission (UGC) test for JRF and Eligibility for Lectureship **(NET-2014)**.
- Member of American Physical Society 2015-2016, 2024-25.

Skills

Theory

- Ultra-cold atoms, Vortex light, Dirac-Fock Theory, Coupled-Cluster Theory, Multiconfigurational Time-Dependent Hartree for Bosons, Atomic structure

Operating System

- Linux & Windows

Programming Language

- Fortran and Python

Application Packages

- MATLAB, Mathematica, GRASP, DALTON, ORIGIN, and Atomic Rydberg calculator.

List of Journal Publications

1. **Anal Bhowmik**, Rhombik Roy, and Sudip Kumar Haldar, Interplay of asymmetry and fragmentation in the many-body tunneling dynamics of two-dimensional bosonic Josephson junctions, **arXiv:2505.01069**.
2. **A. Bhowmik** and D. Blume, Forces on alkali Rydberg atoms due to non-linearly polarized light, **Phys. Rev. A 111, 053120 (2025)**.
3. **Anal Bhowmik** and Ofir E. Alon, Interference of longitudinal and transversal fragmentations in the Josephson dynamics of Bose-Einstein condensates, **New Journal of Physics 26, 123035 (2024).**
4. **A. Bhowmik**, M. Gaudesius, G. Biedermann, and D. Blume, Double, triple, and quadruple magic wavelengths for cesium ground, excited, and Rydberg states, **Phys. Rev. A 110, 043114 (2024).**
5. Sudip Kumar Haldar and **Anal Bhowmik**, Many-body effects in the out-of-equilibrium dynamics of a composite bosonic Josephson junction, **Atoms 12, 66 (2024).**
6. **Anal Bhowmik** and Ofir E. Alon, Coupled-cluster theory for trapped bosonic mixtures, **The Journal of Chemical Physics 160, 044105 (2024).**
7. Swapan Biswas, **Anal Bhowmik**, Arghya Das, Radha Raman Pal, and Sonjoy Majumder, Transitional Strength under Plasma: Precise Estimations of Astrophysically Relevant Electromagnetic Transitions of Ar^{7+}, Kr^{7+}, Xe^{7+}, and Rn^{7+} under Plasma Atmosphere, **Atoms 11, 87 (2023).**
8. Arghya Das, **Anal Bhowmik**, Narendra Nath Dutta, and Sonjoy Majumder, Two-photon polarizability of Ba^{+} ion: Control of spin-mixing processes in an ultracold $^{137}Ba^{+}$-^{87}Rb mixture, **Atoms 10, 109 (2022)**.
9. **Anal Bhowmik**, Narendra Nath Dutta, and Subrata Das, Role of vector polarizability induced by a linearly polarized focused Laguerre–Gaussian light: applications in optical trapping and ultracold spinor mixture, **Eur. Phys. J. D 76, 139 (2022).**
10. **Anal Bhowmik** and Ofir E. Alon, Longitudinal and transversal resonant tunneling of interacting bosons in a two-dimensional Josephson junction, **Scientific Reports 12, 627 (2022)**.
11. **Anal Bhowmik**, Sudip Kumar Haldar, and Ofir E. Alon, Impact of the transverse direction on the many-body tunneling dynamics in a two-dimensional bosonic Josephson junction, **Scintific Reports 10, 21476 (2020)**.
12. **Anal Bhowmik**, Narendra Nath Dutta and Sonjoy Majumder, Vector polarizability of an atomic state induced by a linearly polarized vortex beam: External control of magic, tune-out wavelengths, and heteronuclear spin oscillations, **Phys. Rev. A 102, 063116 (2020)**.
13. Subrata Das, **Anal Bhowmik**, Koushik Mukherjee, and Sonjoy Majumder, Transfer of orbital angular momentum superposition from asymmetric Laguerre-Gaussian beam to Bose-Einstein Condensate, **J. Phys. B: At. Mol. Opt. Phys 53, 025302 (2020)**.
14. Arghya Das, **Anal Bhowmik**, Narendra Nath Dutta, and Sonjoy Majumder, Many-body calculations and hyperfine interaction effect on dynamic polarizabilities at the low-lying energy levels of Y^{2+}, **Phys. Rev. A 102, 012801 (2020)**.

15. **Anal Bhowmik** and Sonjoy Majumder, Tuning of non-paraxial effects of Lagurre-Gaussian beam interacting with two-component Bose-Einstein condensates, **Journal of Physics Communications 2, 125001 (2018)**.
16. **Anal Bhowmik**, Pradip Kumar Mondal, Sonjoy Majumder and Bimalendu Deb, Density profiles of two-component Bose-Einstein condensates interacting with a Laguerre-Gaussian Beam, **J. Phys. B: At. Mol. Opt. Phys 51, 135003 (2018)**.
17. Swapan Biswas, Arghya Das, **Anal Bhowmik** and Sonjoy Majumder, Accurate estimations of electromagnetic transitions of Sn IV for stellar and interstellar media, **Monthly Notices of the Royal Astronomical Society 477, 5605 (2018)**.
18. **Anal Bhowmik**, Narendra Nath Dutta and Sonjoy Majumder, Tunable magic wavelengths for trapping with focused Laguerre-Gaussian beams, **Phys. Rev. A, 97, 022511 (2018)**.
19. Arghya Das, **Anal Bhowmik**, Narendra Nath Dutta and Sonjoy Majumder, Electron-correlation study of Y III-Tc VII ions using a relativistic coupled-cluster theory, **J. Phys. B: At. Mol. Opt. Phys. 51, 025001 (2018)**.
20. **Anal Bhowmik**, Sourav Roy, Narendra Nath Dutta and Sonjoy Majumder, Study of coupled-cluster correlations on electromagnetic transitions and hyperfine structure constants of W VI, **J. Phys. B: At. Mol. Opt. Phys. 50, 125005 (2017)**.
21. **Anal Bhowmik**, Narendra Nath Dutta and Sourav Roy, Precise calculations of astrophysically important allowed and forbidden transitions of Xe VIII, **The Astrophysical Journal 836, 125 (2017)**.
22. Sourav Roy, **Anal Bhowmik**, and Sonjoy Majumder, Precise many-electron calculations of isotope shift for alkali like atoms or ions, **Asian Journal of Physics 25, 1103 (2016)**.
23. **Anal Bhowmik**, Pradip Kumar Mondal, Sonjoy Majumder and Bimalendu Deb, Interaction of atom with nonparaxial Laguerre-Gaussian beam: Forming superposition of vortex states in Bose-Einstein condensates, **Phys. Rev. A 93, 063852 (2016)**.

Book Chapters

1. A. U. J. Lode, O. E. Alon, **Anal Bhowmik**, M. Büttner, L. S. Cederbaum, R. Chitra, S. Dutta, D. Jaksch, H. Kessler, C. Leveque, R. Lin, P. Molignini, L. Papariello, M. C. Tsatsos and J. Xiang, Correlations, Shapes, and Fragmentations of Ultracold Matter, **High Performance Computing in Science and Engineering '22. HPCSE 2022. Springer 2022**
2. A. U. J. Lode, O. E. Alon, J. Arnold, **Anal Bhowmik**, M. Büttner, L. S. Cederbaum, B. Chatterjee, R. Chitra, S. Dutta, C. Georges, A. Hemmerich, H. Keßler, J. Klinder, C. Leveque, R. Lin, P. Molignini, F. Schäfer, J. Schmiedmayer, and M. Zonda, Quantum simulators, phase transitions, resonant tunneling, and variances: A many-body perspective. **International Conference on High Performance Computing in Science and Engineering, Springer, 2021.**
3. A. U. J. Lode, O. E. Alon, M. A. Bastarrachea-Magnani, **Anal Bhowmik**, A. Buchleitner, L. S. Cederbaum, R. Chitra, E. Fasshauer, L. de Forges de Parny, S. K. Haldar, C. Leveque, R. Lin, L. B. Madsen, P. Molignini, L. Papariello, F. Schäfer, A. I. Streltsov, M. C. Tsatsos, and S. E. Weiner, MCTDH-X: The Multiconfigurational Time-Dependent Hartree Method for Indistinguishable Particles High-Performance Computation Project. **High Performance Computing in Science**

and Engineering '20. Springer, Cham.

4. **Anal Bhowmik** and Sonjoy Majumder, Dynamic Polarizabilities and Magic Wavelengths of Sr^+ for Focused Vortex Light, **Quantum Collisions and Confinement of Atomic and Molecular Species, and Photons. Springer Proceedings in Physics 230, 60 (2019).**

Invited talks

1. Frequency-dependent trapping of Rydberg states of cesium, "**Future of Ultracold and Ultrafast Dynamics**" held on Oct 21- Oct 25 2024, The Max Planck Institute for the Physics of Complex Systems (MPIPKS) in Dresden.
2. Frequency-dependent trapping of Rydberg states of cesium, "**Nonequilibrium physics: Spinor BECs and beyond**" held on May 30- May 31 2024, University of Oklahoma, Norman, USA.
3. Double, triple, and quadruple magic wavelengths of Cs ground, excited, and Rydberg states, "**Keck Retreat**" held on April 17- April 18 2024, University of Oklahoma, Norman, USA.
4. Magic wavelengths for Cesium ground and rydberg states in " **Keck Retreat**" held on September 28- September 29 2023, University of Oklahoma, Norman, USA.
5. Superposition of Vortex States in Bose-Einstein Condensates by a non-paraxial Laguerre-Gaussian beam in a workshop named "**Optomechanics, Orbital angular Momentum and Spin-Orbit Interaction of Light**" organized by Raman Centre for Atomic, Molecular and Optical Sciences (RCAMOS) at IACS (12th-13th Jan 2017).

Conferences, Symposiums, and Workshops

1. Delivered a talk entitled " Frequency-dependent trapping of Rydberg states of cesium" in the conference named" **Future of Ultracold and Ultrafast Dynamics**" held on Oct 21–25, 2024, The Max Planck Institute for the Physics of Complex Systems (MPIPKS) in Dresden.
2. Delivered a talk entitled " Frequency-dependent trapping of Rydberg states of cesium" in the conference named" **55th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics**" held on June 3–7, 2024, Fort Worth, Texas USA.
3. Presented a poster entitled "Trapping Rydberg atoms using vortex light" in the **55th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics**" held on June 3–7, 2024, Fort Worth, Texas USA.
4. Delivered a talk entitled "Coupling of longitudinal and transversal fragmentations in the tunneling dynamics of Bose-Einstein condensates" in the conference named" **54th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics**" held on June 5- June 9 2023, Spokane, Washington, USA.
5. Presented a poster entitled "Interference of longitudinal and transversal fragmentations in the Josephson dynamics of Bose-Einstein condensates" in the **"Optical Tweezer"** workshop held on 1st June 2023, University of Oklahoma, Norman, USA.
6. Delivered a talk entitled "Effects of the transverse direction on the many-body tunneling dynamics" in the conference named"**53rd Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics**" held on May 30-June 3 2022, Orlando, Florida, USA.

7. Participated a conference "**From Chaos to Thermalization**" held at Technion, Israel during May 16-19, 2022.
8. Presented a poster entitled "Tunable magic wavelength for trapping with focused optical vortex" in "**International Workshop on Bose-Einstein Condensation and Related Phenomena (IWBECRP)**" held at S. N. Bose National Centre for Basic Sciences, Kolkata, India during 26-28 March, 2018.
9. Presented a poster entitled "Transfer mechanism of angular momentum from non-paraxial optical vortex to atomic BEC" in the conference "**Quantum Optics IX**" organized at the hotel Mercure Gdansk Posejdon, Kapliczna 30, Gdansk, Poland (17th-23rd Sept 2017).
10. Participated in the academic Program "**Quantum Optics and Quantum Technology (QOQT-2017)**", organized by the Department of Materials Science and Raman Centre for Atomic, Molecular and Optical Sciences (RCAMOS) at IACS (5th-8th June 2017).
11. Presented a Poster on the **6th Research Scholar's Day** of the Department of Physics, IIT Karagpur (12th March 2016).
12. Participated in the ICTS School on "**Frontiers in Light-Matter Interactions (ICTS-LMI2014)**" held at Indian Association for the Cultivation of Science, Kolkata (8th-18th Dec 2014).
13. Attended "**Theme Meeting on Ultrafast Science (UFS-2013)**" organized by Department of Physics and Meteorology, IIT Kharagpur and Indian Society for Radiation and Photochemical Sciences, Mumbai (25th–26th Oct 2013).
14. Attended a workshop on "**Introductory Astronomy and Astrophysics**" sponsored by IUCAA, Pune and IUCAA Reference centre, NBU, and organized by the Department of Physics of J. K. College Purulia (23rd Feb-25th Feb 2006).

Names, Addresses, and e-mails of the Collaborators

- **Prof. Doerte Blume**, Homer L. Dodge Department of Physics and Astronomy, University of Oklahoma, USA, Email: Doerte.Blume-1@ou.edu
- **Prof. Ofir E. Alon**, Department of Physics and Haifa Research Center for Theoretical Physics and Astrophysics, University of Haifa, Haifa, Israel, Email: ofir@research.haifa.ac.il
- **Prof. Sonjoy Majumder**, Department of Physics, Indian Institute of Technology Kharagpur, Kharagpur-721302, India, Email: sonjoym@phy.iitkgp.ac.in
- **Dr. Narendra Nath Dutta**, Department of Physics, School of Advanced Sciences, Vellore Institute of Technology (VIT-AP University), Andhra Pradesh-522237, India Email: narendranath-dutta7@gmail.com
- **Prof. Bimalendu Deb**, Department of Material Science, Indian Association for the Cultivation of Science, Jadavpur, Kolkata- 700032, India, Email: msbd@iacs.res.in